Sampling Distributions and the Central Limit Theorem

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Population, sample, parameter and statistic; Normal distribution.
- **Leads to:** Confidence intervals; Hypothesis testing: null and alternative; Z-test and rejection regions; Particle filter.

1. Overview

Key point: Take many samples of the same size, compute each sample's mean, and the means form approximately a normal distribution centred on the population mean, whatever the shape of the **population** (G-1525), provided the samples are large enough and the population has a finite variance.

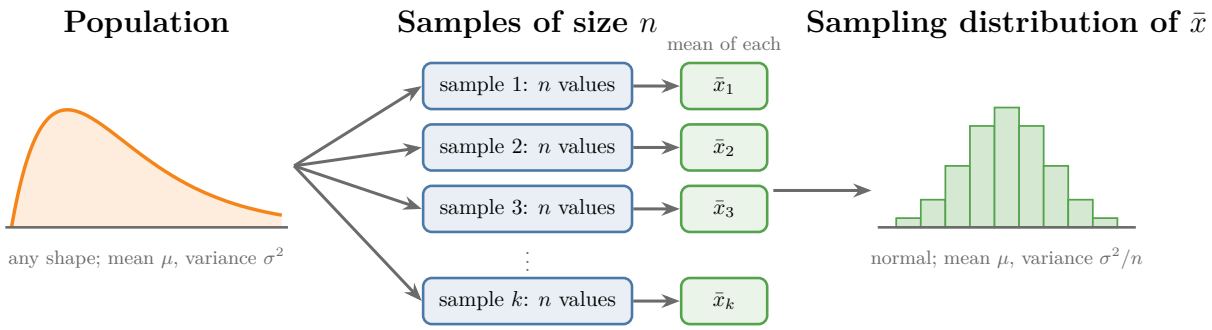


Figure 1: From a population of any shape, through many samples, to the sampling distribution of the mean

Figure 1 shows the process this Note is built on. We draw many **samples** (G-1731) of size n from a population, compute one number per sample (here the mean), and look at the distribution of those numbers. That distribution is a **sampling distribution** (G-1736).

The **central limit theorem (CLT)** (G-364) says what this distribution looks like for the mean: a normal curve with the population's mean and a spread that shrinks as n grows. Here n is the number of values in one sample (50 people, in the salary example below), μ is the population mean, and σ^2 is the population variance, the average squared distance of a value from μ. The CLT holds for skewed, flat and even discrete populations, and it is the base of the next topics, **confidence intervals** (G-446) and hypothesis testing.

This Note covers:

- the sampling distribution of a **statistic** (G-1880);
- the statement of the CLT and its conditions;
- the mean μ and variance σ^2/n of the sample means, and the **standard error** (G-1872);

- simulations from five non-normal populations;
- why the CLT matters.

2. Population and sample, in brief

> **Key point:** We want a number about the whole population, but we can only measure samples.

Recap (see parameters and statistics): the **population** (G-1525) is the entire group we want to study; a **sample** (G-1731) is the part we actually measure. A number computed from the population, such as μ, is a **parameter** (G-1448); the same number computed from a sample, such as $\bar{x}$, is a **statistic** (G-1880).

Our running example is the average monthly salary in India. The population is the salary of all 140 crore people. Asking all of them is impossible, so we ask, say, 50,000 people chosen at random from every state and district, and infer the national average from them.

3. Sampling distributions

> **Key point:** A sampling distribution is the distribution of a statistic (a mean, a variance, a proportion) computed from many independent samples of the same size from one **population** (G-1525).

The salaries of the whole population are a set of numbers, so they follow some distribution. Most likely a skewed one: many people earn little and a few earn a lot, like the orange curve of Figure 1. We do not know its exact shape.

Now we build a **sampling distribution** (G-1736) in four steps:

1. Choose a **sample size** (G-1727), say $n = 50$ people.
2. Draw 50 people at random and record their salaries: **sample** (G-1731) 1.
3. Repeat, drawing a new random sample of 50 each time, until we have, say, 100 samples: 100 sets of 50 numbers.
4. Compute the mean of every sample: $\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_{100}$.

These 100 sample means form the **sampling distribution of the sample mean**. Two sizes must not be confused: the sample size is $n = 50$ (people per sample); the number of samples is 100.

The **statistic** (G-1880) need not be the mean. If we compute the variance of each of the 100 samples instead, the 100 sample variances form the **sampling distribution of the sample variance**. The same works for the standard deviation, the coefficient of variation or a proportion.

A small case with three samples of size $n = 3$ shows the idea in numbers. Each sample gets one mean:

$$\bar{x}_1 = \frac{2+4+9}{3} = \frac{15}{3} = 5$$

$$\bar{x}_2 = \frac{1+3+5}{3} = \frac{9}{3} = 3$$

$$\bar{x}_3 = \frac{6+7+8}{3} = \frac{21}{3} = 7$$

The numbers 5, 3 and 7 are three points of the sampling distribution of the mean. The formal version, for k samples of size n, names the values x_{ij} (the i-th value of sample j; above, $x_{21} = 4$ is the second value of sample 1):

$$\bar{x}_j = \frac{1}{n} \sum_{i=1}^{n} x_{ij}$$

There is one such mean for each sample $j = 1, \ldots, k$.

The symbol $\sum_{i=1}^{n}$ means "add the values for $i = 1$ up to $i = n$". For sample 1 it is:

$$2 + 4 + 9 = 15$$

The sampling distribution is the distribution of $\bar{x}_1, \ldots, \bar{x}_k$; here $k = 3$.

Figure 2 draws this example. Watch each orange mark: it sits at the balance point of its sample's three blue values, and only the marks drop to the bottom line, where the sampling distribution is built.

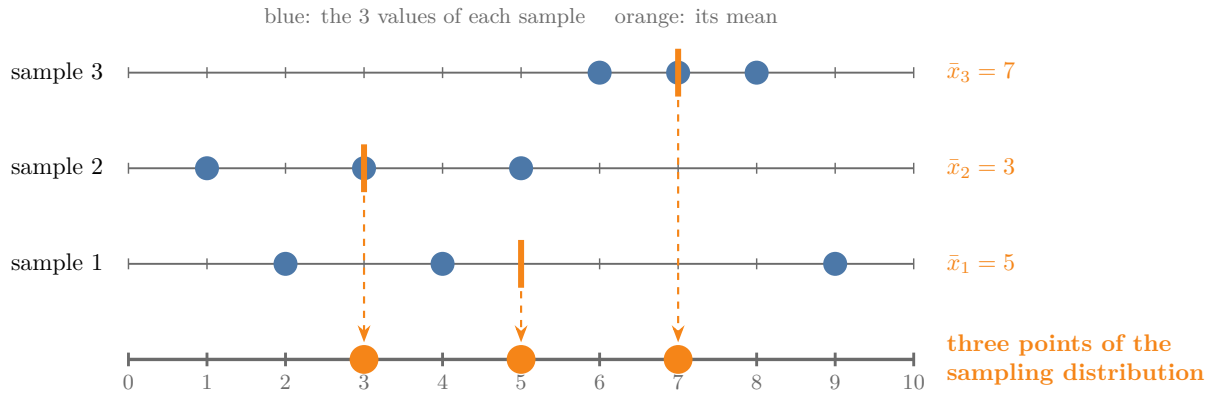


Figure 2: Three samples of size 3 (blue values) and their means 5, 3 and 7 (orange), which become three points of the sampling distribution

3.1 Why sampling distributions matter

> **Key point:** A sampling distribution shows how much a statistic varies from **sample** (G-1731) to sample, and that variability is what inference needs.

Every sample of 50 people gives a slightly different mean. The **sampling distribution** (G-1736) measures exactly how different: it shows the **variability of a sample statistic**. Knowing that variability lets us:

- compute confidence intervals for a population **parameter** (G-1448);
- perform hypothesis tests;
- predict a **population** (G-1525) parameter, such as the mean, from sample data.

The first two are taught in **confidence intervals** (G-446) and the problem hypothesis testing solves. The third is the subject of this Note and of estimating a mean with the CLT.

4. The central limit theorem

> **Key point:** The sample means of large independent samples follow approximately a normal distribution, regardless of the distribution of the **population** (G-1525), as long as its variance is finite.

The idea in plain words, on dice. One die is flat: every face from 1 to 6 is equally likely, and the mean of one die is 3.5. Now roll 30 dice and take the average. The average almost never lands near 1 or 6, because that needs nearly every die to be low, or nearly every die to be high. Most averages land near 3.5, and the further from 3.5, the rarer they get: a bell. Figure 4, further down, tests this with four very different dice.

This is the **central limit theorem** (CLT, G-364). In plain words: the distribution of the **sample** (G-1731) means of many independent and identically distributed values (each value drawn on its own, all from the same population; section 4.1) approaches a normal distribution, regardless of the distribution of the values themselves.

A worked case on a skewed population. An exponential population (most values near 0, a few large) has mean 1 and variance 1. The variance of the sample means is the population variance divided by the **sample size** (G-1727), σ^2/n (section 4.2 shows why). For samples of $n = 30$:

$$\text{variance of the sample means} = \frac{\sigma^2}{n}$$

$$= \frac{1}{30} = 0.0333$$

$$\text{standard deviation} = \sqrt{0.0333} = 0.183$$

So the CLT predicts that the 30-value sample means pile up in a bell centred at 1 with standard deviation 0.183.

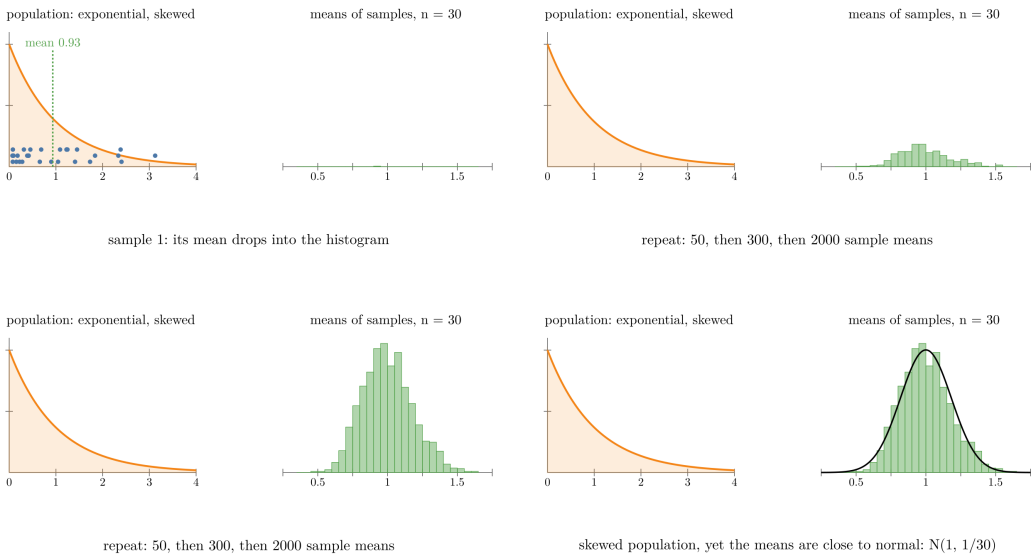


Figure 3: Samples of 30 from a skewed exponential population: each sample's mean drops into the histogram, which grows into the normal curve N(1, 1/30)

Figure 3 runs this experiment, with the numbers above. The population on the left is strongly skewed: most values are near 0 and a few are large. Each sample of 30 values gives one mean, which drops into the histogram on the right. After 2000 samples, the histogram is close to the bell of $N(1, 1/30)$.

The formal version of the prediction, for samples of size n from a population with mean μ and variance σ^2, is:

$$\bar{X} \;\approx\; N\left(\mu,\; \frac{\sigma^2}{n}\right) \quad \text{for large } n$$

Here $\bar{X}$ is the sample mean, and $N(\mu, \sigma^2/n)$ is the normal distribution with mean μ and variance σ^2/n. Check against the worked case, with $\mu = 1$ and $\sigma^2 = 1$:

$$N\left(1,\; \frac{1}{30}\right) = N(1,\; 0.0333)$$

In the salary example of section 3: draw 50 people, record the mean salary, repeat 100 times, and plot the 100 means. The CLT says the plot is approximately a normal curve. The shape of the salaries themselves does not matter. They may be:

- log-normal;
- uniform (every salary in a range equally likely);

- Pareto with a finite variance (α above 2; the Extra on infinite variance below shows what happens otherwise);
- binomial;
- or no named distribution at all.

Figure 4 tests this with four very different dice: a fair one, one that mostly rolls low, a U-shaped one that mostly rolls 1 or 6, and one that mostly rolls 6. For each die we compute the exact distribution of the sum of n rolls. The sums drift to the right and spread out as n grows, so to compare their shapes we re-centre and re-scale each one: subtract its mean $n\mu$ and divide by its standard deviation $\sqrt{n}\,\sigma$ (section 4.2). For a fair die $\mu = 3.5$ and $\sigma = 1.71$, so the sum of 30 rolls has this mean and standard deviation:

$$30 \times 3.5 = 105$$

$$\sqrt{30} \times 1.71 = 9.4$$

In the top row each bar's height is the probability of one face. In the bottom row each bar's height is a probability density (the probability of that value divided by the gap between neighbouring values), so that the bars' area is 1 and they can be laid over the orange $N(0, 1)$ curve. Watch the bottom row: at $n = 1$ the four shapes are as different as the dice; by $n = 10$ they are close to one bell; at $n = 50$ all four sit on the same curve, the **standard normal distribution** $N(0, 1)$ (G-1873; the bell with mean 0 and standard deviation 1). Re-scaling the mean instead of the sum gives exactly the same picture, since the mean is the sum divided by n.

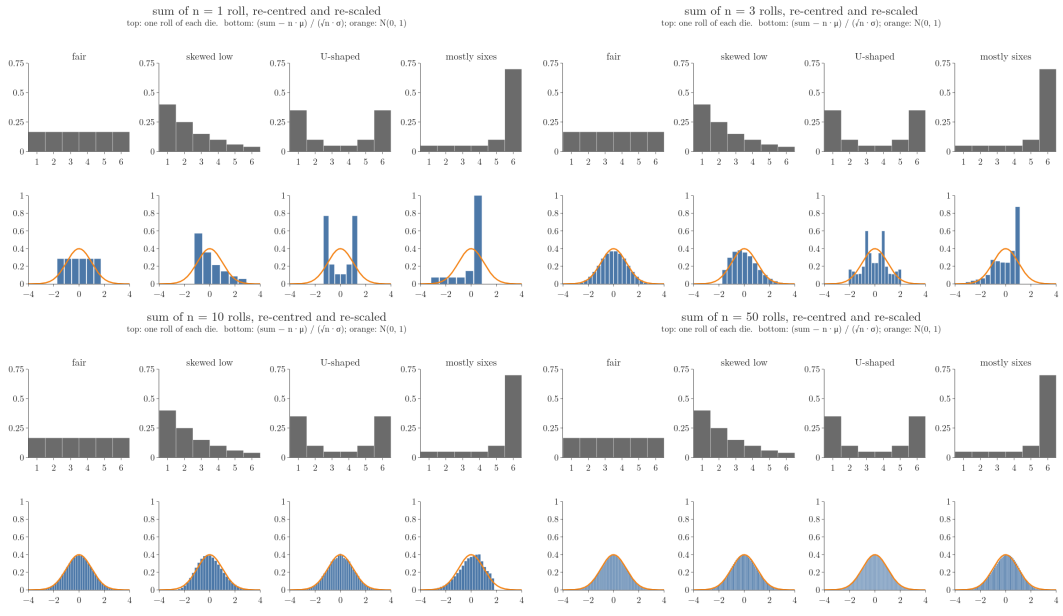


Figure 4: Four different dice (top) and the re-centred, re-scaled sum of n rolls of each (bottom) for n = 1 to 50: all four approach the same normal curve (idea after 3Blue1Brown, "But what is the Central Limit Theorem?")

The CLT makes no claim about the shape of the population or of a single sample. A sample of 100 salaries still looks skewed; only the **means** of many samples form a bell.

> **Extra:** The CLT is a proved theorem, not just an observed fact of nature: Laplace proved a first version in 1810 and Lyapunov a general one in 1901 (Fischer 2011). The intuition: a sample mean adds up many independent pieces, so unusually large and unusually small values tend to cancel, and the sum forgets the shape of each piece. Among shapes with a finite variance, the normal curve is the only one that stays the same when we add copies and rescale (Feller 1971, §VI.1).

4.1 Conditions

> **Key point:** The CLT needs a large enough **sample** (G-1731) (usually $n \geq 30$), a **population** (G-1525) with finite variance, and independent, identically distributed values.

1. **Large enough sample size.** The usual rule is $n \geq 30$, a rule of thumb rather than a sharp limit. How many values are enough depends on the population's shape: a symmetric population gives bell-shaped means already at small n, while a very skewed one (lopsided, with a long tail on one side) needs more (see the Extra box on how fast the means become normal, and section 5.1).
2. **Finite variance.** The population must have a finite variance. Every finite population has one; a theoretical infinite population, such as a Pareto distribution with a small α, may not (see the Extra box on infinite variance below).
3. **Independent and identically distributed (i.i.d.) values. Independent**: one value does not affect another. **Identically distributed**: every value comes from the same population, so each has the same distribution. Drawing at random from one population gives both.

> **Extra:** How fast the means become normal. Skewness (how lopsided a distribution is) and excess kurtosis (how heavy its tails are) are two of the **cumulants**, a family of summary numbers whose first is the mean and whose second is the variance. For independent values the cumulants of a sum add up, just like the variance. Write κ_r for the population's r-th cumulant and $\sigma_{\bar{x}} = \sigma/\sqrt{n}$ for the standard deviation of the sample means (section 4.2). The r-th cumulant of $\bar{X}$ is therefore $n\kappa_r/n^r$, and dividing the third by $\sigma_{\bar{x}}^3$ and the fourth by $\sigma_{\bar{x}}^4$ gives:
>
> $$\sigma_{\bar{x}}^3 = \sigma^3/n^{3/2}$$
>
> $$\sigma_{\bar{x}}^4 = \sigma^4/n^2$$
>
> $$\text{skewness of } \bar{X} = \frac{\gamma_1}{\sqrt{n}}$$
>
> $$\text{excess kurtosis of } \bar{X} = \frac{\gamma_2}{n}$$

where γ_1 and γ_2 are the population's skewness and excess kurtosis. A symmetric population ($\gamma_1 = 0$) gives symmetric means at every n. For the uniform population ($\gamma_2 = -1.2$), $n = 5$ already gives skewness 0.02 and excess kurtosis -0.24 in the Notebook, matching:

$$-1.2/5 = -0.24$$

A very skewed population needs more. The exponential ($\gamma_1 = 2$) still has this skewness at $n = 30$ (section 5.1):

$$2/\sqrt{30} = 0.37$$

Extra: When the variance is infinite, the CLT fails. A Pareto distribution with $\alpha = 1.5$ has mean 3 but infinite variance (see what α does). In 5000 simulated samples, the skewness of the sample means is 17.8 for $n = 30$ and still 19.5 for $n = 300$: no bell appears. For such tails the generalised CLT says the scaled sums approach a skewed "stable" distribution, not a normal one (Gnedenko and Kolmogorov 1954). The Notebook also shows how one value can dominate: in samples of 300, the largest single value makes up at least 55% of the sample's total in 1% of the Pareto samples, against 3.4% at the same point for the exponential population.

4.2 Mean and variance of the sample means

Key point: The **sample** (G-1731) means are centred on the population mean μ, and their variance is σ^2/n; their standard deviation, $\sigma/\sqrt{n}$, is the standard error.

The CLT gives more than the shape. If the **population** (G-1525) has mean μ and variance σ^2:

- the mean of the **sampling distribution** (G-1736) is exactly μ;
- its variance is σ^2/n, where n is the **sample size** (not the number of samples).

The standard deviation of the sampling distribution has its own name, the **standard error** (G-1872) of the mean.

The idea in plain words: bigger samples, tighter means. One die has mean $\mu = 3.5$ and standard deviation $\sigma = 1.71$. Take the average of 100 dice and ask how far it can wander from 3.5:

Step 1. The sum of 100 dice has this mean and standard deviation:

$$\text{mean of the sum} = 100 \times 3.5 = 350$$

$$\text{standard deviation of the sum}$$

$$= \sqrt{100} \times 1.71$$

$$= 10 \times 1.71 = 17.1$$

Step 2. By the CLT the sum is about normal, so 95% of sums land within 2 standard deviations:

$$350 - 2 \times 17.1 = 316$$

$$350 + 2 \times 17.1 = 384$$

Step 3. Dividing the sum by 100 gives the average of the 100 dice. The mean divides too, and the spread divides by 100:

$$\text{mean of the average} = \frac{350}{100} = 3.5$$

$$\text{standard error} = 17.1 \div 100 = 0.171$$

$$\text{check: } \sigma \div \sqrt{n} = 1.71 \div 10 = 0.171$$

$$3.5 - 2 \times 0.171 = 3.16$$

$$3.5 + 2 \times 0.171 = 3.84$$

A single die wanders over the whole range 1 to 6; the average of 100 dice stays within about 0.34 of 3.5 in about 95 percent of tries. A larger sample gives a smaller standard error: four times the **sample size** (G-1727) halves it. So sample means from big samples cluster tightly around μ.

Why the square root. When we add n independent values, the **variances add**, not the standard deviations (Grinstead and Snell 1997, Theorem 6.8). The variance is an average of squared distances, and squaring the distance of a sum gives each value's own square plus cross terms, one value's distance times another's. For independent values, one value's distance says nothing about the other's, so the cross terms average out to 0, and only the separate variances are left. So the sum has variance $n\sigma^2$ and standard deviation $\sqrt{n}\,\sigma$: the spread of a sum grows, but only as $\sqrt{n}$. Dividing the sum by n to get the mean divides that spread by n:

$$\frac{\sqrt{n}\,\sigma}{n} = \frac{\sigma}{\sqrt{n}}$$

The Extra below writes the variance version out.

A second worked case: a gamma population with mean 2 and variance 2, samples of $n = 50$. The standard deviation is:

$$\sigma = \sqrt{2} = 1.414$$

The variance of the sample means is:

$$\sigma_{\bar{x}}^2 = \frac{2}{50} = 0.04$$

$$\sqrt{50} = 7.071$$

$$SE = \frac{1.414}{7.071} = 0.2$$

The sample means vary around 2 with a standard deviation of only 0.2, while single values vary with a standard deviation of 1.414.

The formal version, which both cases follow, is the **standard error** of the mean (the standard deviation $\sigma_{\bar{x}}$ of the sample means):

$$SE = \sigma_{\bar{x}} = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}}$$

The population standard deviation is divided by the square root of the sample size. For the dice the standard error is 0.171 and for the gamma 0.2, as computed above.

> **Extra:** Why the variance is σ^2/n. For independent values, the variance of a sum is the sum of the variances, so $x_1 + \cdots + x_n$ has variance $n\sigma^2$. Dividing by n to get the mean divides the variance by n^2:
>
> $$\mathrm{Var}(\bar{X}) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n}$$
>
> This completes the argument for Bessel's correction in the sample variance. There, measuring distances from $\bar{x}$ instead of μ lost the amount $n(\bar{x} - \mu)^2$. Its average is n times the average of $(\bar{x} - \mu)^2$, which is the variance of $\bar{x}$:
>
> $$n \times \frac{\sigma^2}{n} = \sigma^2$$
>
> That is exactly one σ^2. So the sum of squared distances from $\bar{x}$ averages $(n-1)\sigma^2$, and dividing by $n - 1$ is right on average.

5. Simulating the CLT

> **Key point:** Simulations from uniform, exponential, Poisson, gamma and binomial **populations** (G-1525) all give bell-shaped sample means with the mean and spread the CLT predicts.

We test the theorem by simulation. We pretend a known distribution is the population, draw many **samples** (G-1731) from it, and plot the sample means.

> **Python:** A **sampling distribution** (G-1736) in three lines.
> ```
> import numpy as np
>
> rng = np.random.default_rng(42)      # fixed seed
> samples = rng.uniform(0, 1, size=(1000, 30))
> sample_means = samples.mean(axis=1)
> ```
> `samples` has shape `(1000, 30)`: 1000 samples (rows) of size 30 (columns). `.mean(axis=1)` takes the mean of each row, giving 1000 sample means.

For a **uniform** population on $[0, 1]$ (see the uniform distribution), every value is equally likely: the population is flat, not bell-shaped. Yet the 1000 sample means of size 30 form a bell centred at 0.50, with standard deviation 0.052, matching the CLT's prediction:

$$\sqrt{(1/12)/30} = 0.053$$

With $n = 300$ the bell is narrower: 0.0166, against the predicted 0.0167.

The other populations behave the same way (Figure 5):

- **Exponential**: waiting times between random events; strongly right-skewed, $\mu = \sigma = 1$.
- **Poisson**: counts of events in a fixed interval (see famous distributions); here with mean 3.
- **Gamma**: a right-skewed continuous distribution; here with mean 2 and variance 2.
- **Binomial**: successes in 10 trials with $p = 0.2$ (see the binomial distribution).

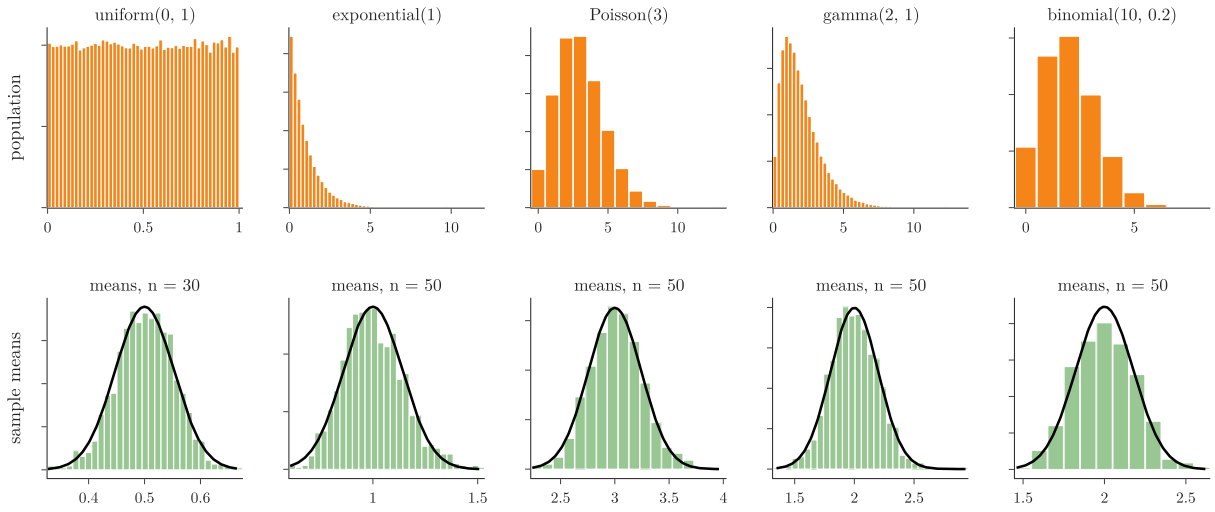


Figure 5: Top: five non-normal populations. Bottom: the means of 1000 samples from each, with the normal curve the CLT predicts

Every population in the top row has a different shape: flat, skewed, discrete. Every histogram in the bottom row is a bell that follows the black normal curve $N(\mu, \sigma^2/n)$.

Population	Skewness of population	Mean of sample means	Std of sample means	CLT: $\sigma/\sqrt{50}$	Skewness of means
exponential(1)	2.00	0.998	0.143	0.141	0.26
Poisson(3)	0.58	2.993	0.249	0.245	0.06
gamma(2, 1)	1.41	2.002	0.195	0.200	0.13
binomial(10, 0.2)	0.47	1.996	0.183	0.179	0.05

The skewness drops from up to 2.00 in the populations to at most 0.26 in the sample means (skewness is a number for how lopsided a distribution is, 0 for a symmetric one; see skewness as distance from the normal shape).

5.1 The effect of the sample size

> **Key point:** As n grows, the **sampling distribution** (G-1736) becomes more normal and narrower; its standard deviation follows $\sigma/\sqrt{n}$.

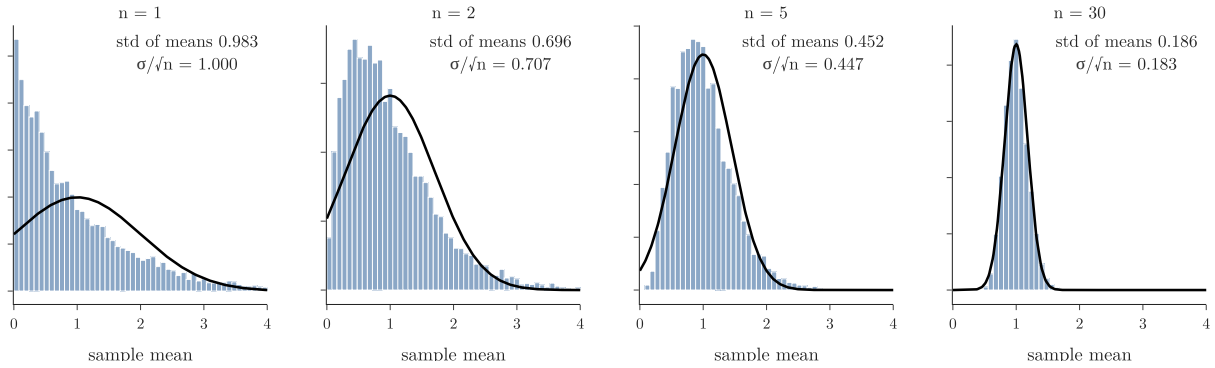


Figure 6: Sample means of an exponential population for n = 1, 2, 5 and 30 (10,000 samples each), with the curve N(1, 1/n)

Figure 6 uses the exponential **population** (G-1525), the most skewed one:

n	Std of sample means	$\sigma/\sqrt{n}$	Skewness of means
1	0.983	1.000	1.93
2	0.696	0.707	1.34
5	0.452	0.447	0.90
30	0.186	0.183	0.38

With $n = 1$, each "mean" is a single value, so the histogram is the population itself. As n grows, the histogram narrows as $\sigma/\sqrt{n}$ predicts, and the skewness falls as $2/\sqrt{n}$ predicts (section 4.1): 2.00, 1.41, 0.89 and 0.37. At $n = 30$ a small skew of 0.38 remains: for very skewed populations, the rule $n \geq 30$ gives a shape that is close to normal, not perfectly normal.

6. Checking the mean and variance

> **Key point:** For a gamma **population** (G-1525) with $\mu = 2$ and $\sigma^2 = 2$, 10,000 **sample** (G-1731) means of size 50 have mean 1.997 and variance 0.0405, against the predicted 2 and 0.04.

To check the second part of the CLT, we use a population whose mean and variance we know exactly: gamma with shape 2 and scale 1, so $\mu = 2$ and $\sigma^2 = 2$. We draw 10,000 samples of size 50.

Quantity	CLT prediction	Simulation
Mean of the sample means	$\mu = 2$	1.9974
Variance of the sample means	$\sigma^2/n = 2/50 = 0.04$	0.0405
Standard error	$\sigma/\sqrt{n} = 0.2$	0.2012

Both properties hold: the sample means are centred on the population mean, and their variance is the population variance divided by n.

Figure 7 replays the check as the samples come in. Watch both curves: after a few samples they swing widely, and after a few hundred they settle on the dashed lines the CLT predicts, 2 and 0.04.

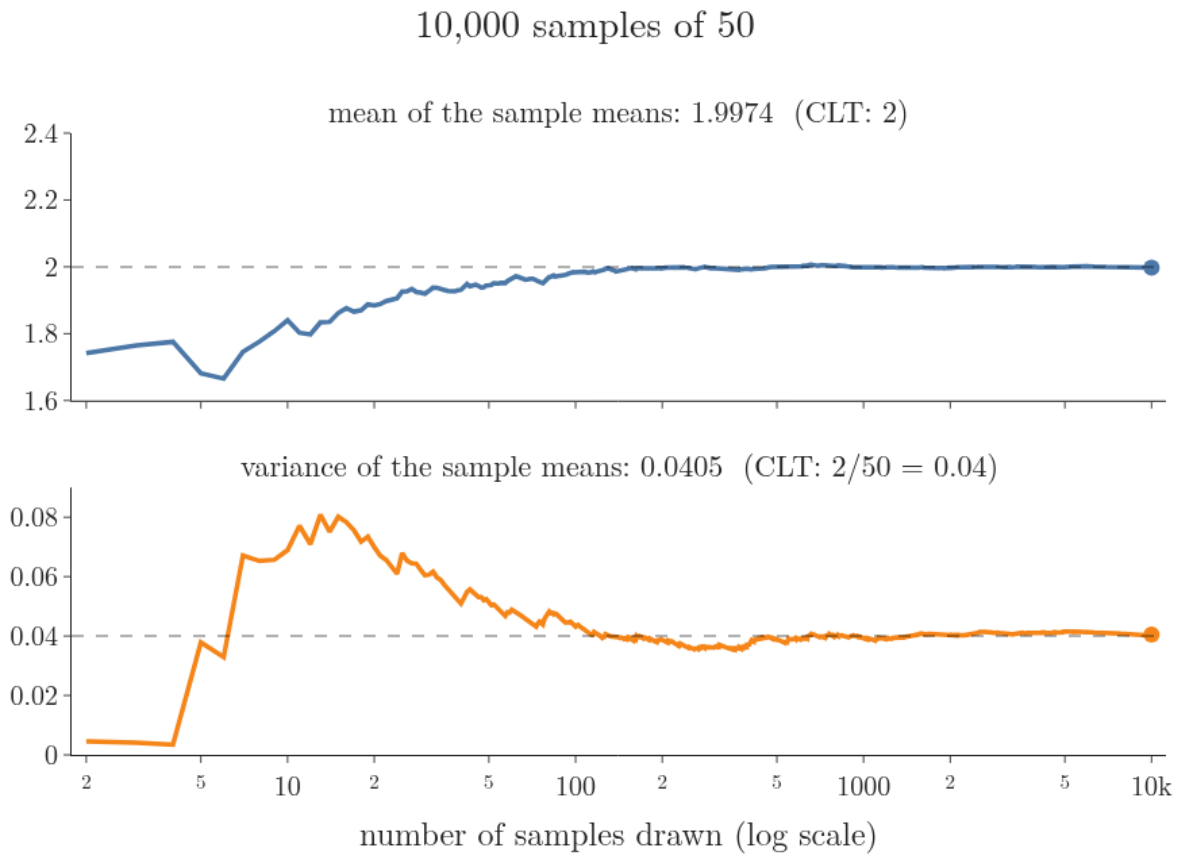


Figure 7: The mean (top) and variance (bottom) of the sample means as 1 to 10,000 samples of 50 accumulate from gamma(2, 1); dashed: the CLT values 2 and 0.04

Python: The check in code.

```
from scipy import stats

gamma = stats.gamma(2)           # shape 2, scale 1
gamma.mean(), gamma.var()       # 2.0, 2.0
rng = np.random.default_rng(7)
means = rng.gamma(2, 1, size=(10_000, 50)).mean(axis=1)
means.mean(), means.var()       # 1.9974, 0.0405
```

7. Why the CLT matters

> **Key point:** The CLT lets us infer a population mean from samples without knowing the **population** (G-1525)'s distribution, and it justifies confidence intervals, hypothesis tests and the tests of linear regression.

The two parts of the theorem turn into a method. We do not know the salary distribution of 140 crore people, but we know that **sample** (G-1731) means are normal around the true mean μ, with **standard error** (G-1872) $\sigma/\sqrt{n}$. So a normal curve built from sample data tells us where μ probably lies: a range that catches μ in, say, 95 percent of samples, called a **confidence interval** (see the standard normal distribution and **confidence intervals** (G-446)). In principle, the variance of the sample means also reveals the population variance:

$$\sigma^2 = n \times \mathrm{Var}(\bar{X})$$

The same works anywhere. A t-shirt company that does not know the heights of all its customers can sample a few thousand of them and estimate the average height of all of them. The same is done on real data in a sampling distribution of the mean fare.

The CLT also justifies many standard techniques:

- **confidence intervals**: a range for a population **parameter** (G-1448);
- **hypothesis tests**, such as the **t-test** (G-1940) (does a sample mean differ from a claimed value?) and **ANOVA** (G-203) (do the means of several groups differ?);
- **linear regression**: its tests on the coefficients rely on normality, which the CLT provides for large samples (Wooldridge, ch. 5).

The CLT has one big condition hidden in the word "random": the samples must be random and representative (see sampling noise and sampling bias). Salaries collected only in cities give a **sampling distribution** (G-1736) centred on the average city salary, not on India's average. The CLT cannot fix a biased sample.

8. Summary

Idea	Formula	Example	Why it matters
Sampling distribution of the mean	distribution of $\bar{x}_1, \ldots, \bar{x}_k$	100 means of samples of 50 salaries	shows how much a sample mean varies from sample to sample, which inference needs
CLT, shape	$\bar{X} \approx N(\mu, \sigma^2/n)$	exponential, $n = 30$: $N(1, 1/30)$	the normal shape holds whatever the population's shape, so normal tables apply to sample means
Mean of the sample means	μ	gamma: 1.9974 against 2	the means centre on the true μ, so one sample mean is an aimed guess at μ
Variance of the sample means	σ^2/n	$2/50 = 0.04$; simulated 0.0405	larger samples give means closer to μ
Standard error	$\sigma/\sqrt{n}$	$1.414/\sqrt{50} = 0.2$	the typical distance of a sample mean from μ, the width used by confidence intervals and tests

- A sampling distribution comes from many **samples** (G-1731) of the **same size**; do not confuse the **sample size** (G-1727) n with the number of samples, because the spread $\sigma/\sqrt{n}$ depends on n, not on how many samples are drawn.
- The CLT holds for any **population** (G-1525) with finite variance, given i.i.d. values and a large enough n (usually $n \geq 30$), so a very skewed population needs a larger n than a symmetric one.
- Larger samples give narrower, more normal **sampling distributions** (G-1736), because the standard deviation of the means is $\sigma/\sqrt{n}$.
- The CLT does not fix biased samples: they must be random and representative, because city-only salaries centre the means on the city average, not on India's.
- This answers the opening question: means of large samples form approximately a normal curve centred on μ whatever the population's shape, so we can infer μ without knowing that shape.

9. Sources

Built from

- CampusX, "Session 43 - Central Limit Theorem | DSMP 2023", YouTube, https://www.youtube.com/watch?v=-WmJDYBor7c
- StatQuest with Josh Starmer, "The Central Limit Theorem, Clearly Explained!!!", YouTube, https://www.youtube.com/watch?v=YAlJCEDH2uY
- Khan Academy, "Sampling distribution of the sample mean", YouTube, https://www.youtube.com/wat KE
- 3Blue1Brown, "But what is the Central Limit Theorem?", YouTube, https://www.youtube.com/watch?

Other references

- Feller, W. (1971). *An Introduction to Probability Theory and Its Applications*, Vol. II, 2nd ed. Wiley. §VI.1.
- Fischer, H. (2011). *A History of the Central Limit Theorem: From Classical to Modern Probability Theory*. Springer.
- Grinstead, C. M. and Snell, J. L. (1997). *Introduction to Probability*, 2nd ed. American Mathematical Society. §6.2, Theorem 6.8 (the variance of a sum of independent random variables is the sum of their variances).
- Gnedenko, B. V. and Kolmogorov, A. N. (1954). *Limit Distributions for Sums of Independent Random Variables*. Addison-Wesley.
- Wooldridge, J. M. *Introductory Econometrics: A Modern Approach*. Cengage. Chapter 5, "Multiple Regression Analysis: OLS Asymptotics".

10. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Sampling distribution (G-1736)	The distribution of a statistic (such as the mean) computed from many independent samples of the same size from one population; it shows how much that statistic varies from sample to sample.
Central limit theorem (G-364)	For large enough samples, the means of many samples follow a normal distribution centred on the population mean, whatever the shape of the data (if its variance is finite); this justifies confidence intervals and hypothesis tests on means.
Standard error (G-1872)	How much a statistic, such as the sample mean, changes from one sample to the next, so how precise it is as an estimate. It is the standard deviation of the sampling distribution; for the mean, $\sigma/\sqrt{n}$.

Term	Meaning
Sample size (n) (G-1727)	The number of values in one sample, written n, such as 50 people; a larger n makes the sample mean vary less from sample to sample (its spread is $\sigma/\sqrt{n}$).
Gamma distribution (G-822)	A family of right-skewed continuous distributions with a shape and a scale parameter.
Independent and identically distributed (i.i.d.) (G-933)	Values that do not affect each other and all come from the same distribution.
Sampling distribution of the sample mean (G-1735)	The distribution of the means of many samples of size n; by the central limit theorem it is close to normal, centred on the population mean, with spread $\sigma/\sqrt{n}$.
Population (G-1525)	The entire group of individuals or objects we want to study.
Sample (G-1731)	The part of a population that we actually measure, such as 50,000 people asked about their salary instead of everyone in India; we study it because measuring the whole population is usually impossible, and use it to draw conclusions about the population.
Confidence interval (G-446)	A range of values, computed from a sample, that we expect to contain a population parameter such as the mean, so it shows how uncertain the estimate is. The method captures the true value in a stated share of repeated samples, such as 95%.
Statistic (G-1880)	A number computed from a sample, such as the sample mean $\bar{x}$; it is used as an estimate of the matching number for the whole population (a parameter).
Parameter (of a function) (G-1448)	A named setting passed to a function, like `sep=";"`.
Standard normal distribution (G-1873)	The normal distribution with mean 0 and standard deviation 1, written $Z \sim N(0,1)$; any normal variable can be converted to it with z-scores, so one table of probabilities serves every normal distribution.
T-test (G-1940)	A hypothesis test about means that uses the sample standard deviation and Student's t-distribution.
ANOVA (G-203)	A test of whether several groups have the same mean (analysis of variance).

Term	Meaning
Exponential distribution (G-733)	The pattern of waiting times between random events: short waits are common and long waits rare (a right-skewed continuous distribution).